

Trigonometric Survival Sheet

Pythagorean Theorem

$$a^2 + b^2 = c^2$$

Unit Circle Key Values

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$
0	0	1	0
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2}$	1	0	undefined

Radians and Degrees

$$180^\circ = \pi \text{ radians}$$

$$\text{degrees} \cdot \frac{\pi}{180}, \quad \text{radians} \cdot \frac{180}{\pi}$$

Signs by Quadrant

QI: All, QII: Sine, QIII: Tangent, QIV: Cosine

Reciprocal Identities

$$\begin{aligned} \sin \theta &= \frac{1}{\csc \theta}, & \cos \theta &= \frac{1}{\sec \theta}, & \tan \theta &= \frac{1}{\cot \theta} \\ \csc \theta &= \frac{1}{\sin \theta}, & \sec \theta &= \frac{1}{\cos \theta}, & \cot \theta &= \frac{1}{\tan \theta} \end{aligned}$$

Pythagorean Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Quotient Identities

$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$$

Cofunction Identities

$$\begin{aligned}\sin\left(\frac{\pi}{2} - \theta\right) &= \cos \theta, & \cos\left(\frac{\pi}{2} - \theta\right) &= \sin \theta \\ \tan\left(\frac{\pi}{2} - \theta\right) &= \cot \theta, & \cot\left(\frac{\pi}{2} - \theta\right) &= \tan \theta \\ \sec\left(\frac{\pi}{2} - \theta\right) &= \csc \theta, & \csc\left(\frac{\pi}{2} - \theta\right) &= \sec \theta\end{aligned}$$

Even-Odd Identities

$$\begin{aligned}\sin(-\theta) &= -\sin \theta, & \cos(-\theta) &= \cos \theta \\ \tan(-\theta) &= -\tan \theta, & \cot(-\theta) &= -\cot \theta \\ \sec(-\theta) &= \sec \theta, & \csc(-\theta) &= -\csc \theta\end{aligned}$$

Sum and Difference Formulas

$$\begin{aligned}\sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \\ \cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B \\ \tan(A \pm B) &= \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}\end{aligned}$$

Double Angle Formulas

$$\begin{aligned}\sin(2\theta) &= 2 \sin \theta \cos \theta \\ \cos(2\theta) &= \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta \\ \tan(2\theta) &= \frac{2 \tan \theta}{1 - \tan^2 \theta}\end{aligned}$$

Half Angle Formulas

$$\begin{aligned}\sin^2 \theta &= \frac{1 - \cos(2\theta)}{2}, & \cos^2 \theta &= \frac{1 + \cos(2\theta)}{2} \\ \tan^2 \theta &= \frac{1 - \cos(2\theta)}{1 + \cos(2\theta)}, & \tan \theta &= \pm \sqrt{\frac{1 - \cos(2\theta)}{1 + \cos(2\theta)}}\end{aligned}$$

Product-to-Sum Formulas

$$\begin{aligned}\sin A \sin B &= \frac{1}{2}[\cos(A - B) - \cos(A + B)] \\ \cos A \cos B &= \frac{1}{2}[\cos(A - B) + \cos(A + B)] \\ \sin A \cos B &= \frac{1}{2}[\sin(A + B) + \sin(A - B)]\end{aligned}$$

Sum-to-Product Formulas

$$\sin A + \sin B = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$$

$$\sin A - \sin B = 2 \cos \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)$$

$$\cos A + \cos B = 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$$

$$\cos A - \cos B = -2 \sin \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)$$

Inverse Trigonometric Functions

$$\sin^{-1}(x), \quad \cos^{-1}(x), \quad \tan^{-1}(x)$$

$$\sin^{-1}(x) : \left[-\frac{\pi}{2}, \frac{\pi}{2} \right], \quad \cos^{-1}(x) : [0, \pi], \quad \tan^{-1}(x) : \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$$

Graphing Trig Functions

General Forms:

$$y = A \sin(Bx - C) + D, \quad y = A \cos(Bx - C) + D, \quad y = A \tan(Bx - C) + D$$

Parameters:

- Amplitude: $|A|$ (height from midline)

- Period:

$$\text{Sine/Cosine: } \frac{2\pi}{B}, \quad \text{Tangent: } \frac{\pi}{B}$$

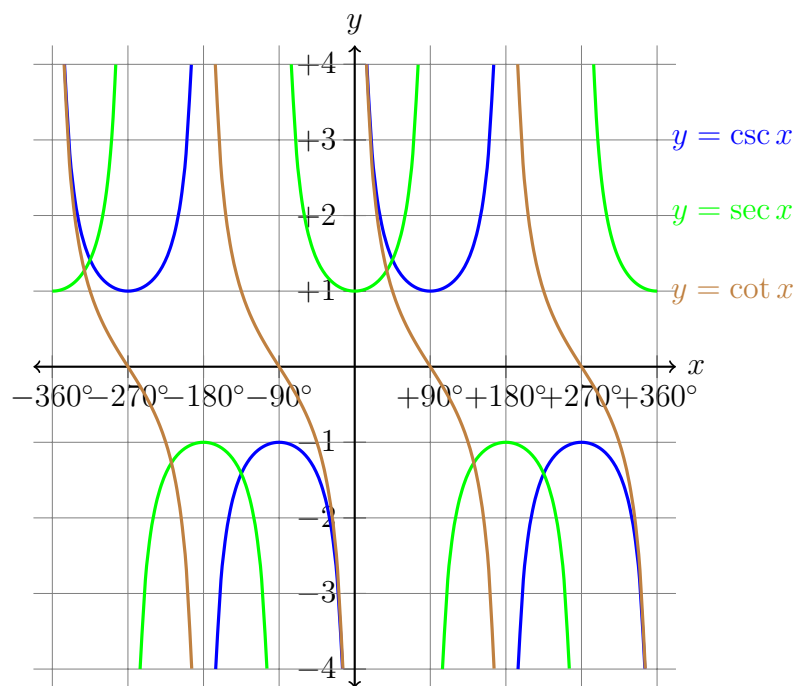
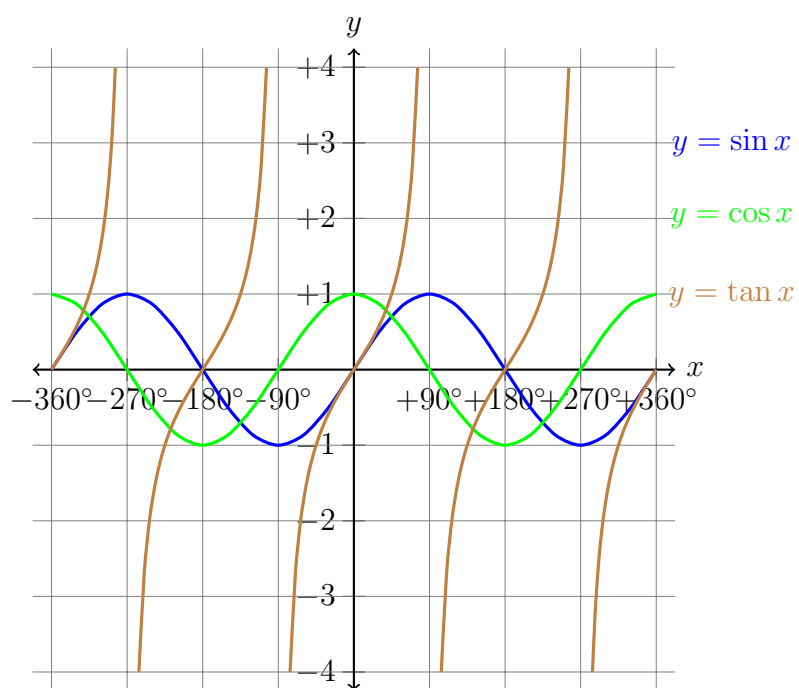
- Phase Shift: $\frac{C}{B}$ (horizontal shift)

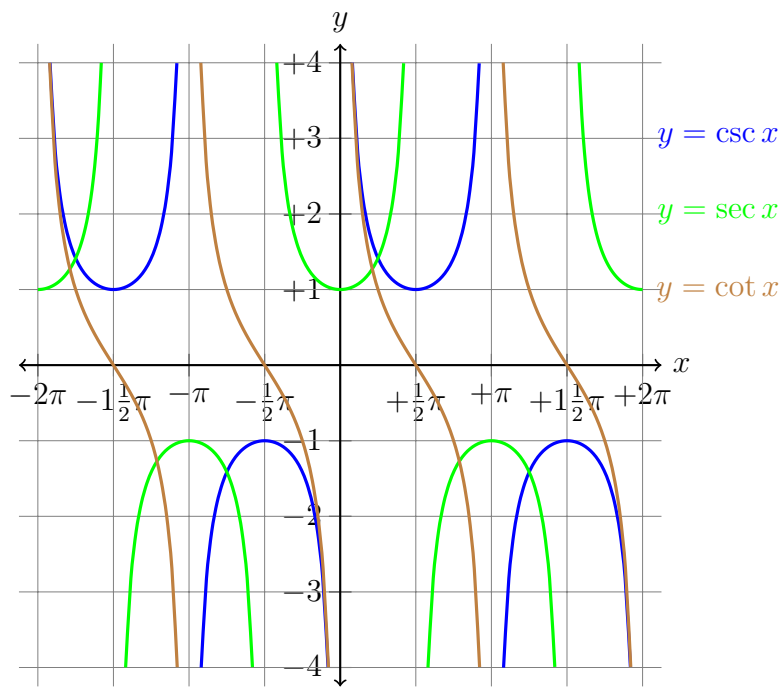
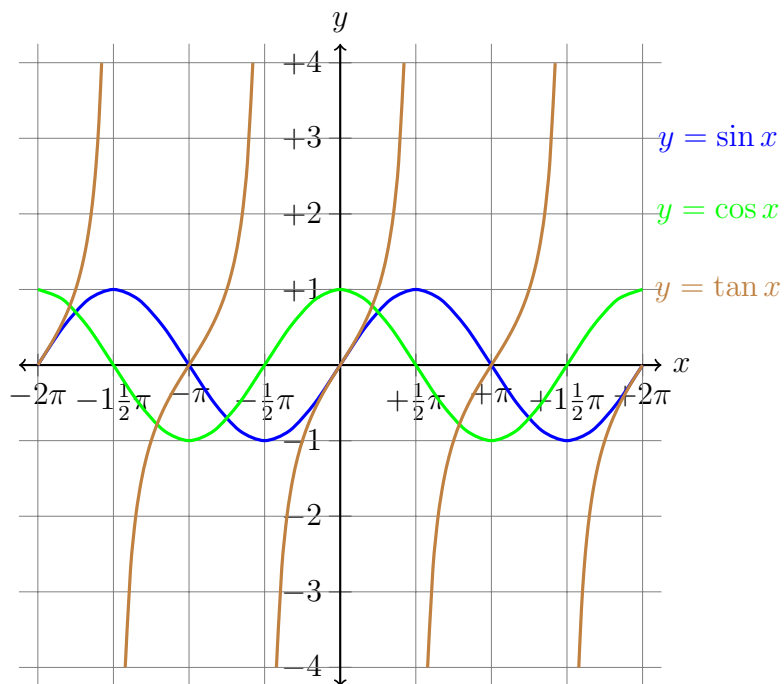
- Vertical Shift: D (moves graph up/down)

Example Key Points:

Function	Period	Amplitude
$y = 3 \sin(2x)$	π	3
$y = 2 \cos\left(x - \frac{\pi}{4}\right)$	2π	2
$y = \tan(x)$	π	undefined

Trig Function Sketches





$$-2\pi - \frac{3}{2}\pi - \pi - \frac{1}{2}\pi \quad 0 + \frac{1}{2}\pi + \pi + \frac{3}{2}\pi + 2\pi$$

$$y = \sin x \quad y = \cos x \quad y = \tan x$$